

BLAKE R. BUCHANAN

SENIOR ROBOTICS ENGINEER

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EDUCATION

Carnegie Mellon University

Master of Science in Robotics

School of Computer Science, Robotics Institute

Pittsburgh, PA

August 2020

University of North Carolina at Charlotte

Bachelor of Science in Mechanical Engineering

Department of Mechanical Engineering and Engineering Science

Charlotte, NC

May 2018

EXPERIENCE

Neya Systems

Senior Robotics Engineer

Boston, MA

January 2024 - Present

- Authoring technical roadmaps and development plans for Neya's in-house off-road simulator, coordinating feature prioritization across autonomy, simulation, and program teams.
- Leading development of Neya's Virtual Integration and Simulation Environment (VISE), a high-fidelity Unreal Engine 5 simulator for off-road autonomous systems in unstructured environments.
- Introduced GPU-based acceleration pipelines in Neya's simulation platform, e.g., ray tracing for high-fidelity lidar simulation, achieving 2.5x FPS improvement over CPU lidar computations.
- Developed and validated product-focused, test-driven software for vehicle-agnostic off-road autonomy, including field integration and testing on physical vehicles.
- Co-developed a hierarchical VLM-based planner grounded in real-time 3D metric-semantic scene graphs, enabling confidence-driven embodied question answering across novel home and office environments with higher success rates and fewer planning steps than prior state-of-the-art approaches.

Sarcos Robotics

Autonomy and Motion Planning Engineer II

Pittsburgh, PA

January 2023 - January 2024

- Led development of an embedded admittance control system with real-time constraints, enabling contact-sensitive manipulation and compliance for human-in-the-loop tasks.
- Developed motion plans and autonomy features for ROS2-based manipulation solutions deployed in aviation, solar construction, and defense applications.
- Implemented real-time C++ software for embedded Linux, including high-level autonomy and control modules for RE2/Sarcos custom manipulators, and conducted routine deployment and testing on hardware.

RE2 Robotics

Software Engineer I

Pittsburgh, PA

October 2021 - January 2023

- Developed a full software stack for robotic manipulation of heavy ammunition rounds within specified time and space requirements for the U.S. Army Applications Laboratory.
- Built software for autonomous manipulation of photovoltaic modules for solar field construction, meeting defined time requirements.
- Led development of motion plans and high-level autonomous behaviors for a mobile manipulation platform performing semi-autonomous casualty extraction from hazardous environments.

The Robotics Institute at Carnegie Mellon University

Biorobotics Lab, Researcher Assistant

Pittsburgh, PA

May 2018 - August 2021

- Researched geometric methods for accomplishing control and coordination for biologically inspired multi-robot systems in dynamic environments, focusing closely on how mechanics govern interactions between robots.
- Developed software in Python, Julia, and MATLAB to test and validate dynamical systems and control strategies.
- Co-authored two published papers concerning fundamental mechanical models for biologically inspired robotic systems and contributed to an awarded NSF research proposal concerning multi-agent robotic systems and interactive mechanics (see here).

University of North Carolina at Charlotte

Scott Kelly's Lab, Undergraduate Research Assistant

Charlotte, NC

May 2016 - May 2018

- Designed experiments and motion control electronics packages for biologically inspired terrestrial and aquatic robots using Arduino microcontrollers.
- Developed an affordable RTK-based differential positioning Raspberry Pi package to track the position of biologically inspired terrestrial and aquatic robots.

TECHNICAL SKILLS

Languages: C++, Python, MATLAB

Tools: Docker, ROS2, CMake, Conan, Unreal Engine, Git, Isaac Lab, MuJoCo

PROJECTS

Training TARS Locomotion via RL in MuJoCo (GitHub)

Language: Python

- Trained locomotion policies for a custom primitive-geometry robot in MuJoCo Playground, implementing PPO and SAC agents via Brax across 2,048 parallelized JAX/MJX environments on CPU.
- Built end-to-end RL infrastructure including a custom MJCF model, reward shaping for forward velocity and stability, checkpoint management with auto-resume, W&B experiment tracking, and a real-time policy visualization pipeline.

GraphEQA: Using 3D Semantic Scene Graphs for Real-time EQA (Website)

Language: C++, Python

- Developed interfaces for and integrated Google's Gemini, Anthropic's Claude, and Meta's Llama 4 into GraphEQA's framework for embodied question answering (EQA).
- Integrated and deployed GraphEQA on Hello Robot's RE2 Stretch Platform for operation in home environments.
- Introduced robust visual logging for debugging scene graph generation, mapping, and inference for simulation and on-robot deployments of GraphEQA.

Swimming In Potential Flow: Fish-like Swimming in a Two Dimensional Fluid (GitHub)

Language: Julia

- Implemented two-dimensional fluid simulation of a flexible fish robot in point vortex flows using Julia.

PUBLICATIONS

Saumya Saxena, **Blake Buchanan**, Chris Paxton, Peiqi Liu, Bingqing Chen, Narunas Vaskevicius, Luigi Palmieri, Jonathan Francis and Oliver Kroemer, (2025) "GraphEQA: Using 3D Semantic Scene Graphs for Real-time Embodied Question Answering" *Conference on Robot Learning (CoRL)*, (PDF)

Blake Buchanan, Tony Dear, Scot David Kelly, Matthew Travers, Howie Choset, (2021) "The Geometric Structure of Externally Actuated Planar Locomoting Systems in Ambient Media," *arXiv.org Preprint*, (PDF)

Blake Buchanan (2020) "Mechanics and Control of Coupled Interactions in Ambient Media," *Master's Thesis*, Carnegie Mellon University, Pittsburgh, PA. (PDF)

Blake Buchanan, Matthew Travers, Howie Choset, and Scott David Kelly (2020) "Stability and Control of Chaplygin Beanie Coupled to a Platform through Nonholonomic Constraints," *ASME DSCC 2020* (PDF)

Tony Dear, **Blake Buchanan**, Rodrigo Abajian-Guerrero, Scott David Kelly, Matthew Travers, and Howie Choset, (2020) "Locomotion of a multi-link nonholonomic snake robot with passive joints," *International Journal of Robotics Research* (PDF)

ADDITIONAL EXPERIENCE

University of North Carolina at Charlotte

January 2016 - May 2018

Department of Mechanical Engineering, Undergraduate Teaching Assistant

- Delivered supplemental lectures for undergraduate dynamics courses, resulting in an overall increase in comfort with course material and performance. Assisted students in learning the PTC Creo CAD package.

The Robotics Institute at Carnegie Mellon University

Pittsburgh, PA

Biorobotics Lab, Robotics Intern

May 2017 - August 2017

- Designed and developed a robot contributing to published research concerning underactuated snake robot control (PDF).